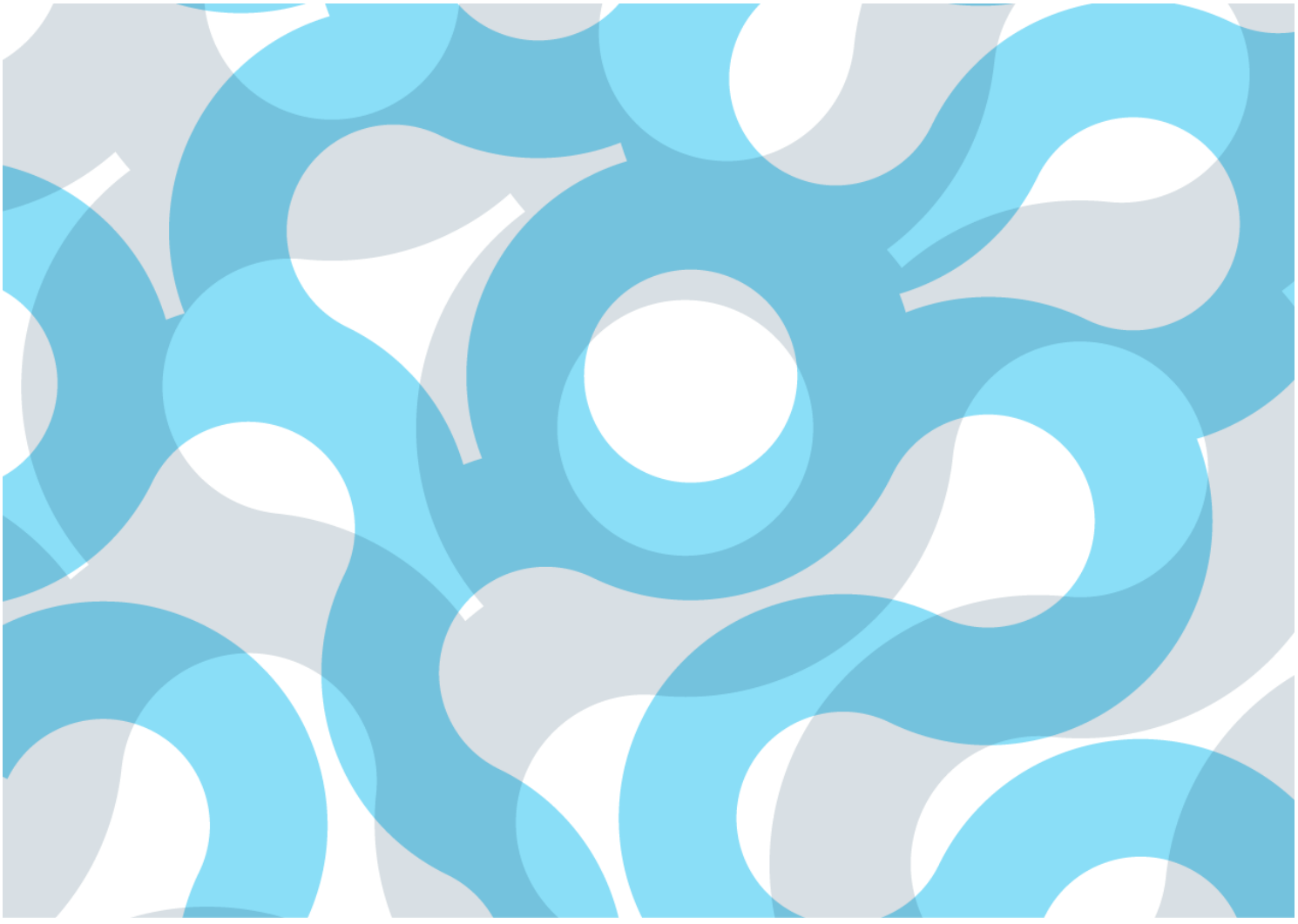
Aladdin Model Documentation

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aladdin by BLACKROCK

Introduction to Portfolio Volatility

Financial Modeling Group | Proprietary and Confidential



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PART I

PORTFOLIO VOLATILITY

Introduction to Portfolio Risk

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Portfolio Volatility

Since Markowitz first introduced mean-variance analysis in 1952, portfolio volatility has been the primary measure of market risk. Even after the development of more sophisticated measures that more accurately capture tail risk (e.g. VaR), risk and performance are usually reported in terms of volatility and Sharpe Ratio.

Part of the reason we still equate risk with volatility is that unlike tail risk measures, volatility is simple to estimate, easily understood, and highly persistent and forecastable. Tail risks are almost by definition unforecastable since they happen so infrequently no model can be calibrated to predict them.

Volatility as a risk measure is especially prevalent on the buy-side, where portfolio managers generally do not rely heavily on derivatives, and where portfolio returns are measured at long enough time intervals (e.g. monthly) that returns begin to approximate a normal distribution.

For all asset classes, we impose the assumption that security returns follow a linear factor structure. That is, an *n*-vector of security returns $\boldsymbol{r}$ can be written as

$$\boldsymbol{r} = \boldsymbol{a} + \boldsymbol{B}\boldsymbol{f} + \boldsymbol{\epsilon}, \quad where\ E(\boldsymbol{f}'\boldsymbol{\epsilon}) = \boldsymbol{0} \tag{1.1}$$

where $\boldsymbol{a}$ is deterministic, $\boldsymbol{f}$ is a *k*-vector of factor returns, $\boldsymbol{B}$ is a *n*x*k* matrix of factor exposures, and $\boldsymbol{\epsilon}$ is a *n*-vector of idiosyncratic returns.[1]

The return of the portfolio of n securities can be written as

$$r_P = \boldsymbol{w}'\boldsymbol{r}, \tag{1.2}$$

where $\boldsymbol{w}$ is the *n*-vector of portfolio security weights.[2]

Portfolio variance is defined as

$$\sigma_p = \left(\boldsymbol{w}'\boldsymbol{E}(\boldsymbol{r}\boldsymbol{r}')\boldsymbol{w}\right)^{1/2} = \left(\boldsymbol{w}'\left(\boldsymbol{B}\mathbf{E}(\boldsymbol{f}\boldsymbol{f}')\boldsymbol{B}' + \mathbf{E}(\boldsymbol{\epsilon}\boldsymbol{\epsilon}') + 2\mathbf{E}(\boldsymbol{f}'\boldsymbol{\epsilon})\right)\boldsymbol{w}\right)^{1/2} = \left(\boldsymbol{w}'\boldsymbol{B}\boldsymbol{\Sigma}\boldsymbol{B}'\boldsymbol{w} + \boldsymbol{w}'\boldsymbol{\Omega}\boldsymbol{w}\right)^{1/2} \tag{1.3}$$

[1] The deterministic component **a** represents the horizon rate of return (HROR) where all risk factors are hold constant except for the passage of time. HROR is described in more detail in the *Parametric Returns* chapter.

[2] In the case where a portfolio has a benchmark, active risk is computed by substituting $\boldsymbol{w}_p - \boldsymbol{w}_b$, portfolio minus benchmark weights, into 1.2.

where Σ is the factor covariance matrix and Ω the block-diagonal covariance matrix of idiosyncratic risk. The last equality holds since the cross-term between factor returns f and idiosyncratic returns ϵ is zero by the orthogonality assumption in equation 1.1.

Covariance Matrix Estimation

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The covariance matrix is central to almost all portfolio risk analytics in the Aladdin Risk Model. A covariance matrix estimate is necessary and sufficient for forecasting volatility, empirical betas, and Parametric VaR, and conducting implied stress testing. Filtered Historical VaR and Monte Carlo VaR also use the covariance matrix as an input to simulate joint factor realizations. While computing a covariance matrix is trivial, the amount of academic literature devoted to estimating more accurate, responsive, and robust matrices is vast.

The Aladdin Risk Model provides users a great deal of flexibility in estimating covariance matrices since the Aladdin risk engine computes most covariance matrices on the fly. This flexibility is a double-edged sword since it requires users to have a better understanding of the Aladdin Risk Model methodologies if users venture too far afield from the default covariance matrix settings.

This chapter primarily provides an overview of the key covariance matrix parameters that can be changed: The observation window, the exponential decay weighting scheme for historical data, the data frequency (i.e. daily, weekly, monthly, etc.), and the choice between overlapping and non-overlapping data. The default settings users can choose from are also listed.

The appendix describes the BFRE (BlackRock Fundamental Risk for Equity model) covariance matrix. The BFRE model covariance matrix is relegated to the appendix since it has a bespoke covariance structure very different from all other models.

Factor covariance estimation

In the simplest case of equal weighting, the factor covariance matrix $\hat{\Sigma}$ can be estimated using the sample covariance matrix

$$\hat{\Sigma} = \frac{1}{N}\sum_{n=0}^{N-1} \boldsymbol{f_{t-n}} \boldsymbol{f'_{t-n}} \tag{2.1}$$

where the returns to k factors is represented by the kx1 vector $\boldsymbol{f_{t-n}}$ and N is the length of the estimation window.

In order to annualize this covariance matrix, the matrix in equation 2.1 is multiplied by 252/q where q refers to the periodicity of returns in days. For example, if the covariance matrix is constructed from weekly returns, the matrix in 2.1 is annualized by multiplying by 252/5.

Notice that the sample mean is not subtracted from the factor returns in equation 2.1. In periods where markets exhibit strong trending, subtracting the sample mean from returns will introduce a severe downward bias in the estimation of variance (see Figlewski [1]).

Following industry practice, covariance is computed directly from second moments where sample means are assumed to be zero. Excluding the sample mean results in a slightly conservative risk forecast. Since the factor mean does not need to be estimated in equation 2.1, the divisor is N rather than N-1.

The following sections will go into more detail on special topics concerning calibration of the matrix, i.e. weighting the data, corrections for asynchronicity, and default values for different parameters of the matrix.

Weighting of historical data

EXPONENTIAL DECAY WEIGHTING

The Aladdin Risk Model by default weights historical observations using an exponentially weighted moving average (EWMA). EWMA puts more weight on the most recent observations in the observation window. Smaller λ in 2.2 corresponds to smaller weights on older observations.

$$\hat{\Sigma} = \frac{1-\lambda}{1-\lambda^N}\sum_{n=0}^{N-1} \lambda^n \boldsymbol{f_{t-n}} \boldsymbol{f'_{t-n}} \tag{2.2}$$

Using l'Hôpital's rule, one can show that as $\lambda \to 1$, equation 2.2 reduces to 2.1.

Instead of defining λ directly, it is usually more convenient to express the decay rate in terms of half-life. The *half-life* $t_{1/2}$ is the time elapsed for the weight on historical observations to decay to half the weight on the most recent observation

$$t_{1/2} = \frac{\log(1/2)}{\log(\lambda)}. \tag{2.3}$$

For instance, a half-life of 8 days means the weight on squared returns 8 days ago is given half the weight of squared returns today. Such a small half-life quickly decays to zero. Setting too short of a half life will result in a

volatility estimate that fluctuates wildly over time due to purely random noise. When using a 252 day window and a half-life of 8 days, the effective number of data points is much smaller than 252 since so little weight is put on the older observations.

The effective number of data points is computed as the inverse of the herfindahl index of the set of decay weights, where the weights are renormalized to sum to one.

$$n_{eff} = \frac{(1-\lambda^N)^2}{(1-\lambda)^2 \sum_{n=0}^{N-1} \lambda^{2n}} \approx \frac{(1+\lambda)}{(1-\lambda)}, \quad where\ N \gg t_{1/2}. \tag{2.4}$$

Combining equation 2.3 and 2.4, a half-life of 8 days with a 252 day window is equivalent to having only 23 data points.[1]

ALTERNATIVE WEIGHTING SCHEMES

The Aladdin Risk Model supports several alternatives to EWMA in covariance estimation. When volatility suddenly spikes, for instance, EWMA may be inappropriate since market conditions going forward are unlikely to be like the past. Instead of EWMA, we might want to estimate a covariance matrix using all periods where VIX was greater than some threshold. The Aladdin Risk Model includes regime-weighted covariance estimation to support this kind of use case.

As a second example, when doing implied stress testing, it usually makes sense to construct the covariance matrix using historical observations that are similar to the hypothetical scenario shocks. For instance, if the scenario envisions a second Greek sovereign debt crisis, it makes sense to weight more heavily data around the first Greek debt crisis when calibrating the covariance matrix. The Aladdin Risk Model supports scenario-weighted covariance estimation just for this purpose. Regime and scenario weighting are described in more detail in the *Stress Testing* chapter.

Asynchronicity

When estimating covariance matrices, we often assume we have observable, frequent and synchronous returns across all assets. In practice, asset returns rarely satisfy these conditions. When returns are not frequent or synchronous, the resulting return volatility and correlation estimates can be severely downwardly biased. This bias can result in higher allocation to sectors with stale prices since those sectors seemingly offer lower volatility and greater diversification. The proximate cause of stale pricing can differ in different contexts, but the underlying cause in all cases is some form of illiquidity. Some causes of stale prices are:

- Asynchronous market closing times. Information that impacts markets do not respect market trading hours.
- Thinly traded securities. Securities that do not trade are often matrix priced by vendors or simply flatlined.
- Artificial smoothing of prices by hedge funds; smoothing of prices by real estate appraisers

[1] Note that the even though the effective number of observations with overlapped data is less than N, the value of N in equation 2.2 is unchanged. If the returns had been subtracted by the sample mean estimates, then the value of N would decrease to n_{eff} defined in equation 2.4. Intuitively, if the returns are highly autocorrelated due to overlap, then demeaning will take out most of the variation around the mean, which will result in a spuriously low estimate of the volatility of the factor return unless the number of observations is reduced.

The simplest work-around to asynchronicity is to select a sampling frequency that is low enough to eliminate most of the bias. For instance, in fixed-income credit markets, practitioners typically measure risk using monthly data due to severely stale pricing caused by the lack of standardization and transparency in OTC markets. In global equity markets, this frequency is typically weekly to account for different closing times in markets. In large cap equity, FX, or futures markets, market frictions are small enough that the optimal sampling frequency may be as high as every 5 minutes. The Aladdin Risk Model allows users to choose daily, weekly, or monthly sampling frequencies to estimate the covariance matrix.

Using a low sampling frequency introduces problems of its own. Using weekly or monthly data means there are less observations to estimate the matrix. The fewer the number of observations, the more noisy the covariance matrix estimates.

In addition, the covariance matrix will be stale between low frequency updates. The covariance matrix will only be current on the day of the matrix update. Further, instead of smoothly updating each day, a low frequency matrix will be flatlined most days and suddenly jump on the day of the covariance matrix update. This jump can make it difficult to disentangle changes in the markets or portfolio from the mechanical process of updating of the matrix. [2]

OVERLAPPING COVARIANCE MATRIX

To mitigate these difficulties, the Aladdin Risk Model introduces overlapping data to simultaneously address the problems of non-synchronous data and the problem of an infrequently updated matrix. An overlapping matrix updates daily using overlapping lower frequency returns.

For instance, instead of a single 5 day non-overlapping return, one can construct returns corresponding to Mon-Mon, Tues-Tues, Wed-Wed returns, etc. Instead of using 52 non-overlapping weekly returns, this operation produces 252 overlapping weekly returns to compute the covariance matrix. By introducing overlapping, the number of effective data points in the estimator increases, lowering sampling variance. The covariance matrix also changes daily, avoiding the sudden jumps in volatility that are introduced when the matrix is only updated infrequently.

The overlapped covariance matrix is computed as

$$\hat{\boldsymbol{\Sigma}} = \frac{1-\lambda}{L(1-\lambda^N)} \sum_{n=0}^{N-1} \lambda^n \sum_{l=0}^{L-1} \boldsymbol{f_{t-n-l}} \sum_{l'=0}^{L-1} \boldsymbol{f'_{t-n-l'}} \tag{2.5}$$

where L is the *overlap parameter*. In our overlapping weekly example, L=5. When L=1, equation 2.5 reduces to 2.2.[3]

Although the overlapping time-series in 2.5 has N observations, the number of effective observations is much lower than that. For instance, the overlapped weekly returns share four observations of daily returns with the two adjacent overlapping returns; as a consequence, the overlapped returns will be highly autocorrelated. Since adjacent overlapping data share so many observations, there are fewer than N independent observations.

As shown in Müller [2], the number of effective observations in overlapped data can be approximated by

[2] The Aladdin Risk Model also incorporates Newey-West to correct for asynchronicity. Newey-West is currently only available for BFRE (BlackRock Fundamental Equity) risk models. The Appendix describes Newey-West and illustrates its relationship with the overlapping methodology

[3] *NB*: the window of unoverlapped observations needs to increase from N to N+L to compute the N overlapped observations in 2.5.

$$n_{eff} \approx \frac{3NL}{2L^2+1} \tag{2.6}$$

For instance, for overlap 5 with 250 data points, n_{eff} equals $3 * 5 * 250/(2 * 25 + 1) \approx 75$. Compared to the weekly non-overlapping estimator, overlapping has increased the number of effective data points by 50%.

Covariance matrix default settings

The Aladdin Risk Model contains default covariance matrix parameter settings referred to as "VaR Types". Table 2.1 uses them. Users can either choose one of these VaR Types or configure their own weighting scheme. Weekly matrices WKY and WKS use returns computed from Wednesday to Wednesday. If there is a holiday on Wednesday in a given week, Tuesday will be used instead.[4] The MTC covariance matrix is computed using monthly returns that are defined by calendar month ends.[5] DLY and WKS are very similar, except WKS is constructed using non-overlapping weekly returns, while DLY is constructed from overlapping 5 day returns with 5 returns per week.

Table 2.1. Covariance Matrix Default Settings

	Defaults			**Configurable**		
VaR Type	Half-life	Window	Overlap	Half-life	Window	Overlap
DLY	40 days	252 days	5	✓	✓	✓
WKS	8 weeks	104 weeks	1		✓	
WKY	26 weeks	104 weeks	1	✓	✓	
MTC	Equal weight	72 months	1	✓	✓	

BFRE covariance estimation

The BlackRock fundamental equity risk model (BFRE) covariance matrix is computed using the same basic methodology described in this chapter. However, there are three additional features that are only incorporated into the BFRE covariance matrix. First, the correlations and volatilities are estimated using different half-lives. Second, the BFRE covariance matrix uses the Newey-West method to correct for asynchronicity rather than overlap. Finally, shrinkage is applied to the BFRE matrix which makes the matrix easier to invert and more suitable for optimization.

Since the BFRE covariance matrix is computed only for BFRE factors, the correlation between non-BFRE risk factors and BFRE risk factors is imposed to be zero to preserve positive-definiteness of the covariance matrix. The BFRE covariance matrix should therefore only be used for portfolios that are close to 100% equity.

[4] *NB*: If a holiday falls on a business day other than Wednesday, weekly returns are still computed Wednesday to Wednesday, which means some weekly returns will have less 5 days.

[5] *NB*: The start and end date of a monthly return are defined as the last business date of each month.

The sections below will describe each unique feature of the BFRE covariance matrix as well as the default parameters for each BFRE model.

Newey-West estimation

As described above, one way to correct for asynchronicity is to use lower frequency overlapped data. An alternative approach is to still use daily data, but explicitly correct for asynchronicity by estimating the degree of autocorrelation and cross-autocorrelation between returns. Newey-West [3] is the most popular method to used to correct for autocorrelation and heteroskedasticity in factor returns.[6]

Newey-West's original result is a positive semi-definite and consistent estimator $\hat{\mathbf{\Sigma}}$ for the L-period sample covariance matrix given by

$$\hat{\mathbf{\Sigma}} = \hat{\mathbf{\Omega}}_0 + \sum_{l=1}^{L} \left(1 - \frac{l}{L}\right) \left[\hat{\mathbf{\Omega}}_l + \hat{\mathbf{\Omega}}_l'\right] \tag{2.7}$$

where $\hat{\mathbf{\Omega}}_l$ represents the matrix of lth order sample auto-covariances and cross auto-covariances ($l = 1$ gives the contemporaneous case) and $L - 1$ is the number of lagged covariance matrices.

Overlapping and the Newey-West methodology are shown to be mathematically equivalent under certain conditions in a subsequent chapter.

Robust Estimation

The BFRE models use Ledoit-Wolf Shrinkage [4] to increase the robustness of the covariance estimate. Shrinkage estimators reduce mean-squared error by trading an increase in bias for an even larger decrease in estimation variance. Shrinkage estimators are particularly important in portfolio optimization where noise in the covariance matrix can result in suboptimal asset allocations.

Ledoit-Wolf Shrinkage [4] locates the optimal solution δ^* to the convex linear combination

$$\delta \boldsymbol{F} + (1 - \delta)\hat{\mathbf{\Sigma}} \tag{2.8}$$

where $\hat{\mathbf{\Sigma}}$ is the sample covariance matrix, $\boldsymbol{F}$ is a highly structured estimator, and $0 < \delta \leq 1$ is the shrinkage constant. In the BFRE model, $\boldsymbol{F}$ is set to the identity matrix so all correlations are shrunk towards zero. In practice the optimal shrinkage intensity used in BFRE is small (typically $< 5\%$).

BFRE covariance matrix default settings

The default parameters for the different BFRE models are shown in table 2.2.

[6] In fact, as of March 2017, Newey-West [3] is the 17th most cited paper in the economics literature.

BFRE has separate long and short half-life covariance matrices for the regional and country-specific models. The BFRE global model uses weekly data due to the asynchronicity of markets across the world.

Notice that the correlations in all BFRE models are estimated using a longer half-life and observation window than the volatilities. The longer half-life is used since correlations tend to be slow moving and much more data is required to calibrate N(N-1)/2 correlations parameters than are needed to calibrate just N volatility parameters.

Table 2.2. BFRE Model Covariance Default Settings

	Half-life		**Newey-West**	
	correlation	**volatility**	**N**	**H**
Equity (Daily)	500 Days	250 Days	10 Days	21 Days
Equity (Weekly)	156 Weeks	52 Weeks	2 Weeks	52/12 Weeks

Insights, Suggestions and Tips

HOW TO CHOOSE COVARIANCE MATRIX SETTINGS

Probably the most common client question is what default covariance matrix parameters should be used. Unfortunately, the answer is ”it depends.” In most cases, we are going to assume the only objective is to construct a covariance matrix that produces the most accurate forecasts of portfolio risk over some specified risk horizon h. We define accuracy in forecasting portfolio variance by mean square error

$$MSE\big(w'\hat{\Sigma}_t w, w'\Sigma_{t+h} w\big) = E\big(w'\hat{\Sigma}_t w - w'\Sigma_{t+h} w\big)^2 \tag{2.9}$$

Generally speaking, the risk horizon h over which we want an accurate forecast should correspond to how long one expects to hold the positions in the portfolio. For instance, if the portfolio official returns are computed monthly, a monthly risk horizon may be most appropriate. A macro hedge fund trading daily may have a daily risk horizon.

In the absence of any portfolio context, we assume the primary goal in estimating a covariance matrix is to choose an estimate that is as close to the ’true’ covariance matrix as possible as measured by mean squared error

$$MSE\big(\hat{\Sigma}_t, \Sigma_{t+H}\big) = E\|\hat{\Sigma}_t - \Sigma_{t+h}\|^2 \tag{2.10}$$

where $\hat{\Sigma}$ and Σ are the sample estimate and true, population covariance matrices respectively and $\|\cdot\|$ denotes the Frobenius norm: $\|A\| = \left(\sum_{i,j=1}^{n} |a_{ij}|^2\right)^{1/2}$.

Before making this assumption, however, in the next section we will briefly touch on other use cases where constructing the most accurate covariance matrix may not be optimal.

Covariance matrix use cases and implications for calibration

Portfolio Construction and Optimization: In portfolio optimization, the accuracy of the inverse of the covariance matrix (also known as the precision matrix), matters more than the accuracy of the covariance matrix itself. Constructing the best matrix for optimization introduces its own special set of challenges. For instance, a covariance estimate of the returns to the stocks in the S&P 500 would provide a reasonable estimate of the risk of any portfolio constructed from those stocks, but would be a very poor sample candidate for portfolio construction since there are too many matrix parameters to estimate. The precision matrix would be incredibly unstable and provide a poor forecast of the actual precision matrix.

Imposing Risk Limits: If the covariance matrix is used to set risk limits, a very responsive matrix may be more accurate but may trigger involuntary selling when volatility is high and prices are low. Or if tracking error volatility bands are very tight on a portfolio that closely tracks its benchmark, a responsive matrix may trigger frequent rebalancing and higher transaction costs. A more stable matrix may be preferable even if it is less accurate.

When the cost to understating risk is high: Minimizing the mean square error forecast assumes that understating and overstating risk are equally undesirable. But the loss function may not be symmetric. For example, the cost to a pension fund with a known liability stream of underestimating risk and incurring a large funding shortfall may be unacceptably high. A conservative estimate of risk is often preferred to an unbiased estimate.

Choosing the right matrix: Overview

Recall that our objective is to minimize the mean square error of our covariance matrix forecast. Probably the most fundamental relationship in statistics is the decomposition of mean square error into the sum of variance plus bias squared.[7]

$$MSE(\hat{\theta}) = Var_{\hat{\theta}}(\hat{\theta}) + Bias_{\hat{\theta}}(\hat{\theta}, \theta)^2 \tag{2.11}$$

Choosing covariance matrix parameters comes down primarily to two choices, both of which further simplify to evaluating a bias-variance tradeoff:

1. Sampling frequency 2. How much weight to put on older data

Sampling frequency

Before choosing the sampling frequency of returns, one should determine the appropriate risk horizon. The risk horizon is the time period over which we are most concerned with having an accurate risk forecast. For instance,

[7] The derivation of equation 2.11 is straightforward.

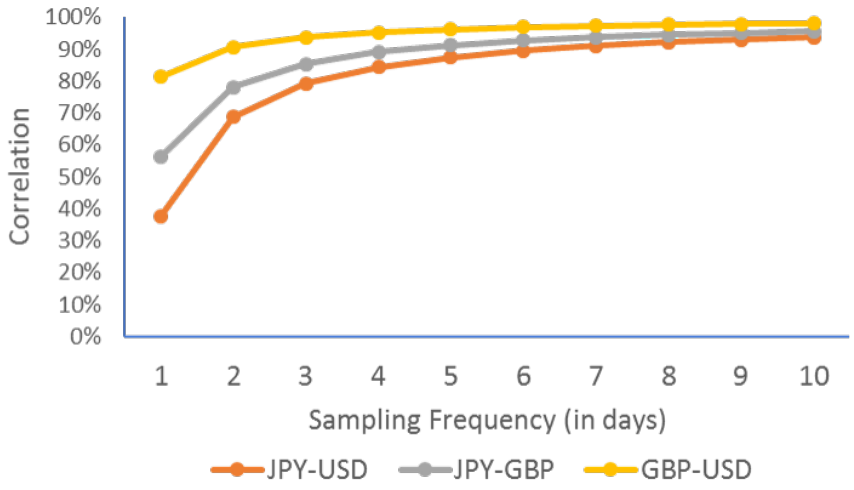
$$MSE(\hat{\theta}) = E_{\hat{\theta}}\left[(\hat{\theta} - \theta)^2\right] = E_{\hat{\theta}}\left[\left(\hat{\theta} - E_{\hat{\theta}}[\hat{\theta}] + E_{\hat{\theta}}[\hat{\theta}] - \theta\right)^2\right]$$
$$= E_{\hat{\theta}}\left[\left(\hat{\theta} - E_{\hat{\theta}}[\hat{\theta}]\right)^2\right] + \left(E_{\hat{\theta}}[\hat{\theta}] - \theta\right)^2 = Var_{\hat{\theta}}(\hat{\theta}) + Bias_{\hat{\theta}}(\hat{\theta}, \theta)^2$$

if the portfolio official returns are computed monthly, a monthly risk horizon may be most appropriate. The risk horizon usually corresponds to how long the investor is expected to hold the position. A macro hedge fund trading daily may have a daily risk horizon.

Given a risk horizon, the first rule in choosing the sampling frequency is that it should be no longer than the risk horizon. In other words, it is never optimal to use weekly returns to calibrate the covariance matrix if the risk horizon is daily.

The choice of sampling frequency comes down to the trade off between variance and bias. Using higher frequency returns lowers the uncertainty (i.e. variance) of the covariance matrix, but comes at the cost of introducing bias. Higher sampling frequencies tend to dampen the empirical correlation between returns[8]

Figure 2.1. Correlation between global markets



Overlapping returns

In most cases, it is preferable to use overlapping returns rather than non-overlapping returns in constructing covariance matrices. Overlapping return matrices make better use of the data since they capture intra-period variation of returns that are missed by non-overlapping data.

- The usage of more data increases the effective number of observations and makes overlapping return matrices more stable through time. However, the increase in stability is somewhat modest. The decrease in volatility of an estimate of standard deviation is only about 20% with overlapped data.
- Overlapping matrices change gradually each day while non-overlapping matrices change only when a new observation is added. If monthly returns are used, there can be a very sudden jump in risk at month end.

Non-overlapped covariance matrices estimated with non-overlapping data have the virtue of being simple to understand and fast to calculate.

[8]Higher sampling frequencies also often dampen the volatility of returns as well.

Window length

The length of the observation window is linked to the choice of half-life. For instance, if the half-life is only ten days, then there is no point having a long observation window since the weights exponentially decay to zero very quickly. On the other hand, one doesn't want to set the observation window too short, since an extreme return that rolls out of the estimation window might cause a spurious change in the matrix today.

Let's stipulate that for a given half-length, we want to set the observation window so that the sum of the weights in the window is 99% of the sum of the weights using an infinitely long history. In other words, the returns data outside the window would have almost no impact on the covariance matrix estimate since it would only get 1% weight if included in the sample.

Given decay parameter λ, the window length that contains 99% of the weight is $\log(0.01/\lambda)$. Assuming a 40 day half-life, this corresponds to a window of 260 days, just slightly longer than a year.

Equivalence of Overlapping and Newey-West

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Equivalence of Overlapping and Newey-West

As shown in 2.5 the overlapping covariance matrix is computed as (in slightly different but equivalent form)

$$\hat{\Sigma}_t^{\mathsf{OL}} = \frac{1-\lambda}{1-\lambda^N} \sum_{n=0}^{N-1} \lambda^n \frac{1}{L} \left(\sum_{l=0}^{L-1} f_{t-n-l} \right) \left(\sum_{l=0}^{L-1} f_{t-n-l} \right)^\top, \tag{3.1}$$

whereas the Newey-West estimator is

$$\hat{\Sigma}_t^{\mathsf{NW}} = \hat{\Omega}_0 + \sum_{l=1}^{L-1} \left(1 - \frac{l}{L}\right) \left(\hat{\Omega}_l + \hat{\Omega}_l^\top\right), \qquad \hat{\Omega}_l = \frac{1}{N} \sum_{n=0}^{N-l-1} f_{t-n} f_{t-n-l}^\top, \tag{3.2}$$

with $\hat{\Omega}_l$ the l-th order auto-covariance. t is the (discrete) time at which we want to calculate the covariance, f_t is the log-factor return at time t, L is the overlap parameter for rolling log-returns with $L = 1$ no overlap, λ is the decay of the smoothing, and N controls the window used for estimation. Note that for $\hat{\Sigma}_t^{\mathsf{OL}}$ we implicitly use $N + L - 1$ returns instead of N.

These two covariance estimates are not equivalent. Clearly, the Newey-West estimator is equally weighted, whereas the overlapping estimator has exponential decay. But even if we let $\lambda \to 1$ such that the overlapping estimator becomes equally weighted, these estimators yield different results. With slight modifications to both estimators, we can achieve equivalence. In the remainder of this section, we use the notation $x^+ = \max\{0, x\}$, $x^- = -\min\{0, x\}$ to split out the positive and negative parts of a real number $x \in \mathbb{R}$.

First, introduce exponential weighing to the Newey-West estimator as follows,

$$\hat{\Sigma}_{\lambda,t}^{\mathsf{NW}'} = \hat{\Omega}_0^\lambda + \sum_{l=1}^{L-1} \frac{\lambda^l - \lambda^L}{1-\lambda^L} \left(\hat{\Omega}_l^\lambda + \hat{\Omega}_l^{\lambda\top}\right), \qquad \hat{\Omega}_l^\lambda = \frac{1-\lambda}{1-\lambda^N} \sum_{n=0}^{N-l-1} \lambda^n f_{t-n} f_{t-n-l}^\top. \tag{3.3}$$

Effectively this is a generalization of the Newey-West estimator, since the original can easily be obtained by letting $\lambda \to 1$.

The modification of the overlapping estimator is two-fold,

$$\hat{\Sigma}_t^{\mathsf{OL}'} = \frac{1-\lambda}{1-\lambda^N} \sum_{n=1-L}^{N-1} \lambda^{n+L-1} \frac{1-\lambda}{1-\lambda^L} \left(\sum_{l=n^-}^{L-1-(N-L-n)^-} f_{t-n-l} \right) \left(\sum_{l=n^-}^{L-1-(N-L-n)^-} f_{t-n-l} \right)^\top. \tag{3.4}$$

First, observe that the normalization by $\frac{1}{L}$ has been replaced with $\frac{1-\lambda}{1-\lambda^L}$. Naturally, we could also have multiplied the Newey-West estimator with $\frac{1-\lambda^L}{1-\lambda}$. However, this would reveal an undesirable bias in the estimator.[1] Second, the limits are adapted to adjust the first and last rolling returns. The mechanics are best illustrated using an example.

[1] TODO:demonstrate this.

Consider the univariate case with $t = N = 4$ and $L = 3$, then

$$\hat{\Sigma}_4^{\mathsf{OL}'} = \frac{1-\lambda}{1-\lambda^4}\sum_{n=-2}^{3}\lambda^{n+2}\frac{1-\lambda}{1-\lambda^3}\left(\sum_{l=n^-}^{2-(1-n)^-} f_{4-n-l}\right)^2 \tag{3.5}$$

$$= \frac{1-\lambda}{1-\lambda^4}\frac{1-\lambda}{1-\lambda^3}\left(\underbrace{\lambda^5\begin{pmatrix}\\ \\ f_1\end{pmatrix}^2 + \lambda^4\begin{pmatrix}f_1\\ +\\ f_2\end{pmatrix}^2}_{\text{no terms on/before } t-N} + \lambda^3\begin{pmatrix}f_1\\ +\\ f_2\\ +\\ f_3\end{pmatrix}^2 + \lambda^2\begin{pmatrix}f_2\\ +\\ f_3\\ +\\ f_4\end{pmatrix}^2 + \underbrace{\lambda^1\begin{pmatrix}f_3\\ +\\ f_4\end{pmatrix}^2 + \lambda^0\begin{pmatrix}\\ \\ f_4\end{pmatrix}^2}_{\text{added partial rolls}}\right). \tag{3.6}$$

Note how the terms on or before $t - N$ have been removed, and partial rolling log-returns have been added that resemble the known parts of the future rolling log-returns. The addition of the leading partial log-returns ensures that the covariance updates quickly with a large covariance in the latest terms, by accounting for these the same L number of times as the older terms. We start depreciating from the head with λ^0.

With these modifications, we have $\hat{\Sigma}_{\lambda,t}^{\mathsf{NW}'} = \hat{\Sigma}_t^{\mathsf{OL}'}$ for $0 < \lambda < 1$ as well as the equally weighted case, $\lambda \to 1$, and for all integer $0 < L < N$.[2] The next section proves the equivalence.

PROOF

For $L = 1$ the proof is trivial, so assume that $L > 1$. Consider the overlap estimator in slightly different notation,

$$\hat{\Sigma}_t^{\mathsf{OL}'} = \frac{1-\lambda}{1-\lambda^N}\sum_{n=1-L}^{N-1}\lambda^{n+L-1}\frac{1-\lambda}{1-\lambda^L}\left(\sum_{l=0}^{L-1} f_t(n+l)\right)\left(\sum_{l=0}^{L-1} f_t(n+l)\right)^\top, \quad f_t(l) = \begin{cases} f_{t-l} & 0 \le l \le N-1, \\ 0 & \text{otherwise}, \end{cases} \tag{3.7}$$

where $f_t(l)$ is the lag function that ensures exclusion of data beyond the current estimation window of length N, by setting everything else to zero.

We will prove that this estimator is equivalent to the exponentially weighted Newey-West estimator,

$$\hat{\Sigma}_{\lambda,t}^{\mathsf{NW}'} = \hat{\Omega}_0^\lambda + \sum_{l=1}^{L-1}\frac{\lambda^l - \lambda^L}{1-\lambda^L}\left(\hat{\Omega}_l^\lambda + \hat{\Omega}_l^{\lambda\top}\right), \qquad \hat{\Omega}_l^\lambda = \frac{1-\lambda}{1-\lambda^N}\sum_{n=0}^{N-l-1}\lambda^n f_{t-n} f_{t-n-l}^\top. \tag{3.8}$$

Effectively this is a generalization of the Newey-West estimator with a Bartlett Kernel as the equally weighted case can be obtained from $\lambda \to 1$.

The proof relies on expanding the interaction of the rolling windows in a smart matter, i.e. expanding the multiplication of the two inner summations. In the univariate case, when we compute $(f_1 + \cdots + f_n)^2$, we get all

[2] For $L = 1$ the Newey-West estimator technically sums from $l = 1$ to 0, which is impossible. In this case we simply replace the sum with zero, so the Newey-West estimator is simply $\hat{\Omega}_0^\lambda$. The equivalence of the two estimators is trivial for $L = 1$.

combinations of indices, e.g. $f_1f_1, f_1f_2, f_2f_1, \ldots$ etc. Instead of summing all combinations, we can also sum over all differences in these combinations. There are n terms where the difference is zero, i.e. $f_1f_1, \ldots, f_nf_n$, $n-1$ terms where the difference is 1 and also $n-1$ where the difference is -1, etc. Mathematically,

$$\left(\sum_{i=0}^{b} f_i\right)\left(\sum_{i=0}^{b} f_i\right)^\top = \sum_{i=0}^{b}\sum_{j=0}^{b} f_i f_j^\top = \sum_{i=-b}^{b}\sum_{j=i^-}^{b-i^+} f_j f_{j+i}^\top, \tag{3.9}$$

where we use the notation $l^+ = \max\{0, l\}$ and $l^- = -\min\{0, l\}$ such that $l^+ + l^- = |l|$ and $l^+ - l^- = l$. Application of this formula gives

$$\hat{\Sigma}_t^{\mathsf{OL}'} = \frac{1-\lambda}{1-\lambda^N}\sum_{n=1-L}^{N-1}\lambda^{n+L-1}\frac{1-\lambda}{1-\lambda^L}\sum_{l=1-L}^{L-1}\sum_{k=l^-}^{L-1-l^+} f_t(n+k)f_t(n+k+l)^\top. \tag{3.10}$$

Further simplification gives

$$\hat{\Sigma}_t^{\mathsf{OL}'} = \sum_{l=1-L}^{L-1}\frac{1-\lambda}{1-\lambda^L}\sum_{k=l^-}^{L-1-l^+}\frac{1-\lambda}{1-\lambda^N}\sum_{n=1-L}^{N-1}\lambda^{n+L-1} f_t(n+k)f_t(n+k+l)^\top \tag{3.11}$$

$$= \sum_{l=1-L}^{L-1}\frac{1-\lambda}{1-\lambda^L}\sum_{k=l^-}^{L-1-l^+}\frac{1-\lambda}{1-\lambda^N}\sum_{n=1-L+k}^{N-1+k}\lambda^{n-k+L-1} f_t(n)f_t(n+l)^\top \tag{3.12}$$

$$= \sum_{l=1-L}^{L-1}\frac{1-\lambda}{1-\lambda^L}\sum_{k=l^-}^{L-1-l^+}\frac{1-\lambda}{1-\lambda^N}\sum_{n=l^-}^{N-1-l^+}\lambda^{n-k+L-1} f_{t-n}f_{t-n-l}^\top. \tag{3.13}$$

The second summation can easily be solved. Since

$$\sum_{k=l^-}^{L-1-l^+}\lambda^{l^- -k+L-1} = \sum_{k=-(L-1-l^+)}^{-l^-}\lambda^{l^- +k+L-1} = \lambda^{(l^- +L-1)}\frac{\lambda^{-(L-1-l^+)} - \lambda^{-l^-+1}}{1-\lambda} = \frac{\lambda^{|l|}-\lambda^L}{1-\lambda}, \tag{3.14}$$

we have

$$\hat{\Sigma}_t^{\mathsf{OL}'} = \sum_{l=1-L}^{L-1}\frac{\lambda^{|l|}-\lambda^L}{1-\lambda^L}\frac{1-\lambda}{1-\lambda^N}\sum_{n=l^-}^{N-1-l^+}\lambda^{n-l^-} f_{t-n}f_{t-n-l}^\top. \tag{3.15}$$

The Newey-West estimator does not include negative lags. Instead it considers the positive lags and their transpose. We write the negative lags as transposed positive lags as follows,

$$\sum_{l=1-L}^{-1}\frac{\lambda^{|l|}-\lambda^L}{1-\lambda^L}\frac{1-\lambda}{1-\lambda^N}\sum_{n=l^-}^{N-1-l^+}\lambda^{n-l^-} f_{t-n}f_{t-n-l}^\top = \sum_{l=1}^{L-1}\frac{\lambda^l-\lambda^L}{1-\lambda^L}\frac{1-\lambda}{1-\lambda^N}\sum_{n=l}^{N-1}\lambda^{n-l} f_{t-n}f_{t-n+l}^\top \tag{3.16}$$

$$= \sum_{l=1}^{L-1}\frac{\lambda^l-\lambda^L}{1-\lambda^L}\frac{1-\lambda}{1-\lambda^N}\sum_{n=0}^{N-l-1}\lambda^n f_{t-n-l}f_{t-n}^\top = \sum_{l=1}^{L-1}\frac{\lambda^l-\lambda^L}{1-\lambda^L}\hat{\Omega}_l^{\lambda\top}. \tag{3.17}$$

The result follows by splitting the positive, negative and zero lag,

$$\hat{\Sigma}_t^{\mathsf{OL}'} = \frac{\lambda^{|0|} - \lambda^L}{1-\lambda^L}\hat{\Omega}_0^{\lambda} + \sum_{l=1}^{L-1}\frac{\lambda^l - \lambda^L}{1-\lambda^L}\hat{\Omega}_l^{\lambda\top} + \sum_{l=1}^{L-1}\frac{\lambda^l - \lambda^L}{1-\lambda^L}\hat{\Omega}_l^{\lambda} = \hat{\Sigma}_t^{\mathsf{NW}'}, \tag{3.18}$$

Q.E.D.

ROLLING ESTIMATION

The primary benefit of the equivalence is that we can construct a Newey-West estimator based on the existing VARServer implementation of the overlap estimator. For implementation purposes, we do need this estimator on a rolling basis. That is, we need to be able to construct the $t+1$ estimate using the estimator at time t, plus some data at the beginning and end of the estimation window.

Using a fairly straightforward split of the summations, we get

$$\begin{aligned}\hat{\Sigma}_{t+1}^{\mathsf{OL}'} = \lambda\hat{\Sigma}_t^{\mathsf{OL}'} &+ \frac{1-\lambda}{1-\lambda^N}f_{t+1}f_{t+1}^{\top} + \frac{1-\lambda}{1-\lambda^N}\sum_{n=1}^{L-1}\lambda^{L-n}\frac{1-\lambda}{1-\lambda^L}\left(\sum_{l=0}^{L-1-n}f_{t-l}f_{t+1}^{\top} + \sum_{l=0}^{L-1-n}f_{t+1}f_{t-l}^{\top}\right) \\ &- \lambda^N\frac{1-\lambda}{1-\lambda^N}f_{t-(N-1)}f_{t-(N-1)}^{\top} - \frac{1-\lambda}{1-\lambda^N}\sum_{n=1}^{L-1}\lambda^{N+n-1}\frac{1-\lambda}{1-\lambda^L}\left(\sum_{l=n}^{L-1}f_{t+1-(N-L)-l}f_{t-(N-1)}^{\top} + \sum_{l=n}^{L-1}f_{t-(N-1)}f_{t+1-(N-L)-l}^{\top}\right).\end{aligned} \tag{3.19}$$

For the case when we have the estimate at t and wish to retrieve the estimate at $t-1$, we rearrange to get

$$\begin{aligned}\lambda\hat{\Sigma}_{t-1}^{\mathsf{OL}'} = \hat{\Sigma}_t^{\mathsf{OL}'} &+ \lambda^N\frac{1-\lambda}{1-\lambda^N}f_{t-N}f_{t-N}^{\top} + \frac{1-\lambda}{1-\lambda^N}\sum_{n=1}^{L-1}\lambda^{N+n-1}\frac{1-\lambda}{1-\lambda^L}\left(\sum_{l=n}^{L-1}f_{t-(N-L)-l}f_{t-N}^{\top} + \sum_{l=n}^{L-1}f_{t-N}f_{t-(N-L)-l}^{\top}\right). \\ &- \frac{1-\lambda}{1-\lambda^N}f_tf_t^{\top} - \frac{1-\lambda}{1-\lambda^N}\sum_{n=1}^{L-1}\lambda^{L-n}\frac{1-\lambda}{1-\lambda^L}\left(\sum_{l=0}^{L-1-n}f_{t-1-l}f_t^{\top} + \sum_{l=0}^{L-1-n}f_tf_{t-1-l}^{\top}\right)\end{aligned} \tag{3.20}$$

Part II

Portfolio VaR

Introduction to VaR

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This is an introduction to the primary risk analytics derived from the Aladdin Risk Model's estimated covariance matrix. This chapter will introduce two of the most important portolio risk measures, portfolio volatility and Value-at-Risk. There are three major VaR models supported in the Aladdin Risk Model: Analytical VaR, Historical VaR, and Monte Carlo VaR.

The two subsequent chapters will go into much more detail into the calibration methodology used for Historical and Monte Carlo VaR calculation.

VaR

For a given probability α, Value-at-Risk (VaR) is defined as the threshold such that the probability that the portfolio return exceeds the threshold return is equal to α. Given a portfolio return cumulative distribution function F, VaR for portfolio P at confidence level α is defined as

$$\text{VaR}_p(\alpha) = -F^{-1}(1-\alpha). \tag{4.1}$$

That is, the α% VaR is computed as the $1-\alpha$ quantile from the historical sample distribution of portfolio returns. Note that by convention, VaR is reported with a positive sign when the $1-\alpha$ quantile return is negative.

$$\text{VaR}_{p1+p2}(\alpha) = \text{VaR}_{p1}(\alpha) + \text{VaR}_{p2}(\alpha) \tag{4.2}$$

Although VaR is one of the most common risk measures, it has the undesirable property that it violates a principal tenet of diversification known as subadditivity. Subadditivity states that the risks of two assets should be not greater than the risk of the sum of the assets.

Expected Shortfall, also known as CVaR (i.e. conditional VaR) satisfies subadditivity as a risk measure. For a given probability α and risk horizon H, Expected Shortfall is defined as the average portfolio loss conditional on the loss exceeding α% VaR

$$\text{ES}_p(\alpha) = \frac{1}{\alpha}\int_{a=0}^{\alpha} VaR_p(a)da = E(-r_p | r_p < \text{-VaR}_p(\alpha)). \tag{4.3}$$

Expected Shortfall is computed as the average return in the historical sample that exceeds the α% VaR confidence level.

VaR Methodologies

The Aladdin Risk Model offers three different methods for computing VaR: Analytical VaR, Historical VaR, and Monte Carlo VaR. This section provides a brief overview and discusses the pros and cons of each method, while the two subsequent chapters discuss Historical and Monte Carlo VaR calculation in more detail.

ANALYTICAL VAR

Analytical VaR, also known as Parametric or Delta-Normal VaR, imposes the assumption that asset returns have linear exposure to a set of risk factors, and that the risk factors are all normally distributed. Given these two assumptions, VaR reduces to a scalar multiple of tracking error volatility. Given normality, the scalar multiplier is the value of the inverse of the standard normal cumulative distribution evaluated at a confidence level α. Analytical VaR is defined as

$$\text{AVaR}_{\alpha}(\sigma_T) = -\Phi^{-1}(1-\alpha)\,\sigma_T \tag{4.4}$$

where $-\Phi^{-1}$ is the inverse cdf of the standard normal distribution. So 95% VaR for a portfolio is simply 1.96 times portfolio volatility since 95% of the density of a normal distribution is less than 1.96.

Analytical VaR often understates risk whenever there is explicit or embedded optionality (i.e. non-linearity) since Analytical VaR only captures first order greeks (e.g. δ,ν). This first order approximation is only reasonable at very short horizons or when options have little non-linear behavior.

Analytical VaR especially understates risk at high confidence levels since asset returns are often fat-tailed and/or skewed rather than normal.

Both Historical VaR and Monte Carlo VaR capture both the non-linearity of portfolios as well as fat-tailed return distributions.

HISTORICAL VAR

Historical VaR is computed by first generating a history of hypothetical portfolio returns using the current portfolio exposure to risk factors. α-Historical VaR is sorting portfolio returns from smallest to largest and selecting the appropriate quantile from the non-parametric distribution.

Historical VaR is defined as

$$\text{HVaR}(\alpha, T) = Percentile\left\{\{r_{p,t}\}_{t=1}^{T}, \alpha\right\} \tag{4.5}$$

Modeling tail events requires a long history of data since by definition events in the tails of the distribution don't happen very often. Computing stable HVaR estimates for small α requires using a long history. Using a long history

has the drawback that HVaR estimates may not be very responsive to current market conditions. An extreme event far in the past but still in the observation window can substantially impact current VaR estimates, a phenomenon known as 'ghosting'.

In order to construct VaR estimates that are more responsive to current market conditions, Hull and White (1998) introduced Filtered Historical VaR which involves filtering (i.e. dividing) security or factor returns by ex-ante volatility. These filtered historical returns are then rescaled back to the current estimate of volatility. Volatility-scaled (i.e. Filtered) historical VaR is defined as

$$\text{Volatility Scaled HVaR}(\alpha, T) = \sigma_T Percentile\left\{ \{ \frac{r_{p,t}}{\sigma_t} \}_{t=1}^{T}, \alpha \right\} \tag{4.6}$$

The Aladdin Risk Model allows users to compute either Historical or Volatility Scaled Historical VaR estimates. Unsurprisingly, volatility scaled historical VaR estimates will track volatility closely through time while historical VaR will be not. Volatility scaled HVaR will usually perform better in backtests, while HVaR will be much more stable through time.

Historical VaR vs Filtered Historical Simulation

It is important not to confuse HVaR with Filtered Historical Simulation (FHS). Historical VaR involves no simulations. Historical VaR depends on the historical realizations of risk factors. By definition, there is only one historical realization of factor returns for each date.

Filtered historical simulation is another technique not supported in the Aladdin Risk Model. In FHS, factor returns are usually filtered using a GARCH process. [1] One then creates N simulations by bootstrapping these filtered factor returns and then rescaling them by the current volatility estimate from the GARCH model. If the FHS is a single period model, the VaR forecast is constructed from the N simulations. If the FHS is multi-period, the bootstrapped return is used to update the GARCH parameters, and another N simulations are bootstrapped from the returns data and the process repeats. In technical terms, HVaR is a single period IGARCH model while FHS is a multiperiod GARCH model.

[1] NB: HVaR filters returns using EWMA (or IGARCH) estimate of volatility

Parametric Returns

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Overview

BlackRock Solutions uses a multi-factor parametric return model to compute ex-ante portfolio risk across all asset classes. The expected return of each asset is a function of sensitivities to broader market movements, known as systematic factors. This document characterizes all factors in the Aladdin Risk Model. We define each factor, provide exposure and return units, and give clarifying examples when appropriate.

Return Methodology

Asset returns are expressed in terms of sensitivities – or exposures – to systematic factors. In this section, we outline basic factor return methodology. We cover details specific to each asset class in Sections 5.3 through 5.6.

DEFINITIONS

We express ex-ante asset returns using a standard linear factor model:

$$R = e'F + \epsilon \tag{5.1}$$

where R is ex-ante return, e is the vector of asset exposures, F is a vector of factor returns, and ϵ is random error not captured by systematic factors (which is on average zero).

Factor returns are expressed either as ratios or absolute differences. Equities, foreign exchange, and other price-based factors are expressed in ratio terms. We assume that prices are lognormally distributed, so the daily return on any price-based factor is computed as follows:

$$f_t = log(\frac{l_t}{l_{t-1}}) \tag{5.2}$$

where l_t is the factor level at time t. Because prices are lognormally distributed, it follows that log returns of price-based factors are normally distributed.

Daily interest rate changes and non-corporate spread changes are expressed in absolute difference terms:

$$f_t = l_t - l_{t-1} \tag{5.3}$$

We assume that arithmetic differences in levels are normally distributed. Because asset returns are a weighted sum of factor returns, it follows that asset returns are assumed to be normally distributed.

Changes in corporate spreads are measured either in ratio terms or in absolute difference terms. For more details, please consult Section 5.3.2 .

LINKING METHODOLOGY

We compute weekly and monthly factor returns by linking daily returns. Factor return over k days is expressed as follows:

$$f_{t,k} = \sum_{i=0}^{k-1} f_{t-i} \tag{5.4}$$

Note that weekly and monthly returns are normally distributed, because they are defined as the sum of normally distributed daily returns.

SIGN CONVENTION

Some factors have an implicit negative sign embedded in their exposure. For instance, when interest rates increase, bonds lose value. Conversely, when equity markets increase, stocks appreciate in value. In both cases, factor levels increase, but the direction of return is fundamentally different.

To distinguish between these two scenarios, each factor has an associated sign convention that is implicitly included in the parametric return calculation.

Fixed Income

INTEREST RATES

The return of a government bond with no credit risk is a function of key rate durations (KRDs) and key rate bucket convexities (KRBCs):

$$R = -\sum_{i=1}^{11} KRD_i * f_i + \sum_{i=1}^{4}\sum_{j=1}^{4} \frac{KRBC_{i,j} * f_i * f_j}{2} \tag{5.5}$$

We provide a further explanation of key rate durations and convexities below.

Key Rate Durations

Key rate durations measure the sensitivity of bond prices to changes in key rates. Collectively, they capture the risk of non-parallel yield curve shifts.

We compute key rate durations by using the Aladdin Analytics Engine. We revalue each bond when a given key rate is shocked up and down by 50 basis points in isolation, and compute the associated first order sensitivity to yield changes. Mathematically, key-rate duration to the ith factor is defined as follows:

$$KRD_i = \frac{V^- - V^+}{2 * V_0 * \Delta y} \tag{5.6}$$

where V_0 is the initial value of the bond, V^- is the bond value when the key rate shifts downwards, V^+ is the bond value when interest rates shift upwards, and Δy is the simulated shift in yield.

Key Rate Convexities

Convexity factors capture second-order sensitivities to interest rate changes. We compute key rate convexities using the following formula:

$$KRBC_{i,j} = V_0 * KRD_i * KRD_j \tag{5.7}$$

SPREAD FACTORS

Corporate Spreads

Credit spread volatility is empirically proportional to current spread level. For example, if a corporate bond's spread doubles, its spread volatility will double as well. This relationship holds not only across time, but also cross-sectionally; if one bond has twice the spread as another, its spread volatility will be twice as high.

To capture this behavior, corporate spread exposure is defined as duration times spread (DxS) and corporate spread returns are typically expressed in percentage change terms:

$$R = -\underbrace{D * S}_{Exposure} * \underbrace{\frac{\Delta S}{S}}_{FactorReturn} \tag{5.8}$$

where R is the return attributable to corporate spreads, D is spread duration, S is the current spread level, and ΔS is the change in spread.

The relationship between spread volatility and spread level breaks down over long horizons. This is because at time horizons longer than three years, spread levels tend to mean-revert to their long run average level. Since credit spread volatility at short horizons is accurately forecasted by percent spread volatility times current spread, one might suspect that the best estimate of credit spread volatility at longer time-horizons starts to converge to percent spread volatility times long run average spread. For this reason, those with longer investment horizons or those who desire a more stable estimate of spread volatility can disable the scaling of spread volatility to current level of spread.

Inflation Spreads

Inflation-linked bonds are mapped to nominal interest rates and breakeven inflation spreads. Inflation factor levels are defined as the difference between real and nominal government bond yields for a given maturity:

$$l_t = Real_t - Nominal_t \tag{5.9}$$

While breakeven inflation spreads are typically expressed as the difference between nominal and real yields, Aladdin inflation factors are expressed with the opposite convention. That is, an increase in factor level corresponds to a decrease in breakeven inflation level.

This sign convention ensures that inflation-linked bonds have positive duration to inflation spreads within Aladdin.

TIME FACTOR (HORIZON RATE OF RETURN)

The time factor captures carry: the deterministic component of return assuming all factors are unchanged over a given time horizon. Because it is deterministic, it is not included in the Value-at-Risk calculation. However, since it affects portfolio return, it is included in stress tests.

Formally, the time factor measures the difference between future and starting values of a security over a given

time horizon, assuming that cash flows paid over the horizon are reinvested and all systematic factors remain unchanged. This is known as the horizon rate of return:

$$HorizonReturn = 100 * \frac{FuturePrice + ReinvestedCashFlows - InitialPrice}{InitialPrice} \tag{5.10}$$

To summarize, the horizon rate of return of a corporate bond is calculated using the following steps:

1. The option adjusted spread (OAS) on the initial date is calculated based on the initial price of the security.
2. The cash flows between the initial and horizon dates are generated, and assumed to be invested at the current risk-free rate.
3. The security is repriced on the horizon date holding both its OAS and the yield curve constant.
4. Equation 5.10 is used to calculate the resulting horizon return.

Note that time factor return is expressed in calendar days, and exposure units of the time factor are in basis points per day.

Equity

Aladdin uses three complementary risk models to measure equity risk: STORM (Security Total Return Methodology), BFRE (BlackRock Fundamental Risk for Equities) and STORM 2.0 (a hybrid of both STORM and BFRE with additional capabilities).

In STORM, each equity security is treated as a separate factor and portfolio risk is calculated using an asset-by-asset covariance matrix of realized returns.

Conversely, BFRE categorizes company fundamentals (such as region, industry, and styles) into factors to impose far more structure in computing equity risk. This section covers the STORM and BFRE risk models in further detail.

STORM

In STORM, each equity security is represented as a separate risk factor. An equity portfolio with *N* equity assets will have *N* factor exposures – one for each equity security in the portfolio.

The STORM return of a portfolio is computed as follows:

$$R = \sum_{i=1}^{N} w_i * r_i \tag{5.11}$$

Above, $w_i$is defined as the weight that the portfolio holds in the security, and r_i is defined as the realized return of the ith security. The time series of STORM portfolio returns can be interpreted as the hypothetical portfolio realized return had an investor constantly rebalanced their equity positions to their current state.

STORM takes an unstructured approach and thus captures the natural evolution of each stock's return time series. The strength of this approach is that it makes fewer assumptions about market structure than BFRE, which

assumes that equity securities have a constant country, sector, and style structure. This unstructured approach is particularly important during times of market change such as the emergence of the internet industry prior to the tech bubble or the surge in junk stocks during the recovery rally of 2009. Furthermore, STORM more directly captures idiosyncratic effects, which might be too narrow or too recent to appear in fundamental factors.

BFRE

BFRE uses company fundamentals, in addition to historical market data, to decompose portfolio risk into a series of factors based on technical and fundamental characteristics. Many of these characteristics are similar to those used by portfolio managers and financial analysts to form investment views on the future performance of companies. These characteristics are based on balance sheet variables, income statements, analyst estimates, country and industry of business, and historical market data. One of the primary advantages of using a fundamental equity risk model is the decomposition of risk using an intuitive set of fundamental equity risk factors that align with how portfolio managers structure their investments.

The model structure includes a market factor, style factors, industry factors, country factors and currency factors:

$$R = 1 * f_{mkt} + \sum_{k=1}^{N} e_{Ind(k)} * f_{Ind(k)} + \sum_{l=1}^{P} e_{Cty(l)} * f_{Cty(l)} + \sum_{j=1}^{M} e_{Sty(j)} * f_{Sty(j)} + \epsilon_i \tag{5.12}$$

Style and industry factors vary by region, reflecting local variation in the importance of different factors.

Currency

SPOT CURRENCY FACTORS

FX factors are expressed as spot returns on foreign currencies from a U.S. investor perspective. FX return of a portfolio can be represented as such:

$$R_{currency} = \sum_{i \in C} w_i * f_i \tag{5.13}$$

where w_i is the weight of the portfolio in currency i, C is the set of currency factors, and f_i is the spot return on the ith currency from a USD perspective.

FACTOR CROSS-CURRENCY

Non-USD investors who wish to express currency risk contributions using their domestic currency can leverage the FX factor cross-currency feature. FX factors can be expressed relative to any user-specified cross-currency:

$$R_{currency} = \sum_{i \in C} w_i * (f_i - f_{xccy}) \tag{5.14}$$

where f_{xccy} is the return on the selected cross-currency, from a USD perspective.

Changing cross-currency is effectively a change in numeraire. That is, individual currency risk contributions may vary, but total currency risk will not change.

This holds for two reasons. First, total currency exposures must sum to zero because a long position in one currency is a short position in another. Second, the f_{xccy} term subtracted from each currency factor f_i is a constant with respect to i. Using these two properties, one can show that the use of any non-USD cross-currency will yield the same total currency risk.

Note that changing factor-cross currency is fundamentally different from changing a portfolio's base currency. The investor is still exposed to the same underlying currency risks, which are simply expressed in a different perspective.

Alternatives

The Aladdin Risk Model covers Private Equity, Real Estate, Infrastructure Equity, Hedge Funds, and Commodity asset classes.

COMMODITIES

The commodity futures risk model constructs constant maturity factors to estimate the portfolio risk contributions of commodity futures holdings. A constant maturity model is designed to capture the term structure of volatility present in commodity futures (e.g. declining volatility with increased time to expiry for hard-to-store commodities).

The model's coverage extends across the major commodity sub-types: Industrial Metals, Precious Metals, Agriculture, Livestock, Energy and Softs. Each commodity sub-type has 12 corresponding constant maturity risk factors. The first risk factor maps to the nearest maturing contract, while the twelfth factor maps to the twelfth nearest contract to maturity.

The current model directly uses the price returns for each of the 12 nearest expiring contracts. Constant maturity factors are constructed by consistently rolling each factor as the nearest contract expires. Each commodity future is directly mapped to its corresponding factor (a mapping that also updates each time the nearest contract expires). That is, this model is the commodity futures-based analogue of the Aladdin STORM (Security Total Return) risk model for publicly listed equities.

PRIVATE EQUITY

The central premise of the private equity model is that public and private equity investments are exposed to many of the same underlying economic risks.

Therefore, to measure private equity risk, we adjust public equity market data for the specific attributes of a private equity investment, such as vintage year, strategy, investment vehicle, regional breakdowns, and sector

allocations. The return of a private equity investment can be expressed as follows:

$$R = \beta_{lvg} * (f_{mkt} + \sum_{k=1}^{N} e_{Ind(k)} * f_{Ind(k)} + \sum_{l=1}^{P} e_{Geo(l)} * f_{Geo(l)} + \sum_{j=1}^{M} e_{Cap(j)} * f_{Cap(j)} + \epsilon_i) \tag{5.15}$$

where β_{lvg} is the leverage of the portfolio. The model includes a market factor, industry factors, geography factors, and capitalization factors. Note that industry, geography, and capitalization factors are constructed net of the market factor.

REAL ESTATE

Private real estate exhibits a strong empirical relationship with its public equivalent, after adjusting for the effects of seasonality and smoothing. The real estate 2.0 risk model maps private real estate investments onto proprietary factors, while incorporating attributes such as property leverage, geography-by-property type exposure and strategy type.

The return of a private real estate investment can be expressed as follows:

$$R = \beta_{lvg} * (\sum_{i=1}^{S} e_{segment(i)} * f_{segment(i)} + \sum_{j=1}^{S} e_{segment(j)} * \epsilon_{private,segment(j)} + \epsilon_{property}) \tag{5.16}$$

where β_{lvg} is the adjustment for property leverage (loan-to-value), $\epsilon_{private,segment(j)}$ is private excess for geography-by-property segment j, and $\epsilon_{property}$ is the model's property-specific return.

HEDGE FUNDS

Portfolio risk in Aladdin is typically modeled using a bottom-up approach in which security exposures are linearly aggregated to form portfolio exposures. This requires position-level information, which is often not available for hedge funds.

BRS uses a performance-based approach in modeling hedge fund exposures. We define a set of hedge fund risk factors designed to capture the return of common hedge fund strategies. We then regress hedge fund return time series onto hedge fund factors to calculate portfolio exposures:

$$R = \sum_{i=1}^{N} \beta_i * f_i \tag{5.17}$$

where β_i is the multiple regression beta to factor f_i. The ability to regress a time series of fund returns onto factors is available interactively through PortfolioRiskTools.

Hedge fund factors are split into primary and secondary categories. Primary factors measure the broad asset class performance of equity, rates, commodities, and currencies. Secondary factors explain intra-asset class performance, such as value vs. growth,momentum, currency carry, or merger arbitrage.

INFRASTRUCTURE EQUITY

Infrastructure is defined as investing in facilities or services focused on the movement of goods, people, water and energy. Aladdin's infrastructure model calculates the economic risk of investments in infrastructure equity. The return of an infrastructure equity investment is modeled as follows:

$$R = 1 * f_{mkt} + \sum_{k=1}^{N} e_{Ind(k)} * f_{Ind(k)} + \sum_{l=1}^{P} e_{Geo(l)} * f_{Geo(l)} + \epsilon_{construction} + \epsilon_{project} \tag{5.18}$$

where f_{mkt} is the market factor, $\epsilon_{construction}$ is private infrastructure specific risk, and $\epsilon_{project}$ is project-specific risk. Note that industry and geography factors in this model are expressed relative to the market factor.

Full Revaluation

Parametric return is a linear, first-order approximation of asset returns. When there are small changes in factor returns, parametric returns approximate realized returns closely.

However, for large market movements – which are often simulated through the use of stress tests – parametric returns may deviate further from realized returns. This is especially true for derivative securities with nonlinear payoffs, such as options.

To address this, BRS has built a full revaluation framework that reprices applicable securities in stress scenarios where market movements are significantly large. In these scenarios, securities are repriced using a bottom-up valuation framework and parametric return is adjusted accordingly.

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